PHYSICS

1) If pressure P, velocity V and time T are taken as fundamental physical quantities, the dimensional formula of force is

(1) PV^2T^2
(2) $P^{-1}V^2T^{-2}$
(3) PVT^2
(4) $P^{-1}VT^2$

2)

Force F and density d are related as $F = \frac{\alpha}{\beta + \sqrt{d}}$, then the dimensions of α are.

(1) $M^{3/2}L^{-1/2}T^{-2}$
(2) $M^{3/2}L^{-1/2}T^{2}$
(3) $M^{3/2}L^{1/2}T^{2}$
(4) $M^{3/2}L^{1/2}T^{-2}$

3) The length of a cylinder is measured with a metre scale having least count 0.1 cm. Its diameter is measured with Vernier callipers having least count 0.01 cm. Given that length is 5.0 cm and diameter is 2.0 cm. The percentage error in the calculated value of the volume will be :

(1) 1%
(2) 2%
(3) 3%
(4) 4%

4) In a new system unit of length, mass and time are 10 cm, 10g and 0.1s respectively then force of 1N will be equal to how much in new system :-

(1) 0.1
(2) 1
(3) 10
(4) 100

5) A wire has a mass of (0.3 ± 0.003)g, radius (0.5 ± 0.005)mm and length (6 ± 0.06)cm then, maximum percentage error in the calculation of density is :-

(1) 1%
(2) 2%
(3) 3%

(4) 4%

6) The potential energy of a particle varies with distance x as $U = \frac{A\sqrt{x}}{x + B}$. The dimensions of AB are :-

(1) $ML^{5/2}T^{-2}$

(2) $M^1L^2T^{-2}$

(3) $M^{3/2}L^{3/2}T^{-2}$

(4) $M^1L^{7/2}T^{-2}$

7) The resultant of the vectors $\vec{A}$ and $\vec{B}$ is perpendicular to the vector $\vec{A}$ and its magnitude is equal to half the magnitude of vector $\vec{B}$.The angle between $\vec{A}$ and $\vec{B}$ is :-

(1) 120°

(2) 150°

(3) 135°

(4) None of these

8) If $\vec{A} = 3\hat{i} + 4\hat{j}$ and $\vec{B} = 6\hat{i} + 8\hat{j}$ and A and B are the magnitudes of $\vec{A}$ and $\vec{B}$, then which of the following is not true ?

(1) $\vec{A} \times \vec{B} = \vec{0}$

(2) $\frac{A}{B} = \frac{1}{2}$

(3) $\vec{A}.\vec{B} = 48$

(4) A = 5

9) In a vernier callipers, 10 divisions of vernier scale coincides with 9 divisions of main scale. one division of main scale is of 0.1 cm. If in the measurement of inner diameter of cylinder zero of vernier scale lies between 1.3 cm and 1.4 cm of main scale and 2nd division of vernier scale coincides with main scale division then diameter will be :-

(1) 1.30 cm

(2) 1.34 cm

(3) 1.32 cm

(4) 1.36 cm

10) Two full turns of Screw gauge cover a distance of 1 mm on main scale. The total number of circular scale divisions are 100. The measured value of thickness of sheet will be :

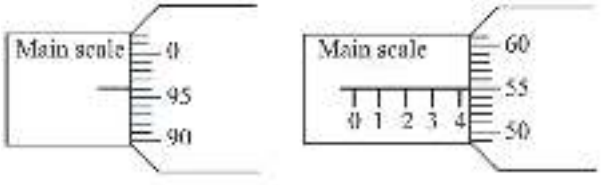


Fig.(1) SCREW GAUGE Fig.(2)

(1) 4.555 mm

(2) 4.295 mm

(3) 2.295 mm

(4) 2.245 mm

11) Two full turns of the circular scale of a screw gauge cover a distance of 1mm on its main scale. The total number of divisions on the circular scale is 50. Further, it is found that the screw gauge has a zero error of –0.03 mm. While measuring the diameter of a thin wire, a student notes the main scale reading of 3mm and the number of circular scale divisions in line with the main scale as 35. The diameter of the wire is :-

(1) 3.32 mm

(2) 3.73 mm

(3) 3.67 mm

(4) 3.38 mm

12) A car moving on a straight road covers one third of a certain distance with speed 20 km/h and the rest with speed 60 km/h. The average speed is :-

(1) 40 km/h

(2) 80 km/h

(3) $46\frac{2}{3}$ km/h

(4) 36 km/h

13) Match the following columns and mark the correct option from the codes given below.

	Column I		**Column II**
(A)	Constant positive acceleration	(p)	Speed may increase
(B)	Constant negative acceleration	(q)	Speed may decrease
(C)	Constant displacement	(r)	Speed is zero
(D)	Constant slope of v-t graph	(s)	Speed must increase
		(t)	Speed must decrease

Codes

	A	**B**	**C**	**D**
(1)	q, r	s	p	t
(2)	p, q	p, q	r	p, q
(3)	p	s, r	s	q
(4)	q	p	t	s, q

(1) 1

(2) 2

(3) 3

(4) 4

14) A particle moves on a straight line, with uniform acceleration –5 m/s^2. Distance travelled by it in three consecutive intervals of 1 sec. are in ratio of 3 : 1 : 1. Then the total distance travelled in three seconds is :

(1) 25 m

(2) 12.5 m

(3) 22.5 m

(4) 20 m

15) Two trains A & B 100 km apart are travelling towards each other on different tracks with starting speed of 50 km/h for both. The train A accelerates at 20 km/h^2 and the train B retards at the rate 20 km/h^2. The distance covered by the train A when they cross each other is :

(1) 45 km

(2) 55 km

(3) 65 km

(4) 60 km

16) A stone is thrown upwards from a tower with a velocity 50 ms^{-1}. Another stone is simultaneously thrown downwards from the same location with a velocity 50 ms^{-1}. When the first stone is at the highest point, the relative velocity of the second stone w.r.t. the first stone is (assume that second stone has not yet reacheadthe ground) :

(1) Zero

(2) 50 ms^{-1}

(3) 100 ms^{-1}

(4) 150 ms^{-1}

17) Trajectory of a particle in projectile motion under gravity with horizontal x axis and vertical y axis is given as:

$y = x - \frac{x^2}{80}$. Here x and y are in meters. Origin is at point of projection. For this projectile motion, with $g = 10 m/s^2$

Column –I		Column –II	
(A)	Angle of projection (in degree)	(P)	20
(B)	Path of projectile	(Q)	80
(C)	Maximum height (in meter)	(R)	45°

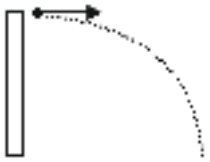
(D)	Horizontal range (in meter)	(S)	parabolic

(1) (A) - P; (B) - Q; (C)- R; (D)- S

(2) (A) - R; (B) - S; (C)- P; (D)- Q

(3) (A) - P; (B) - Q; (C)- S; (D)- R

(4) (A) - R; (B) - S; (C)- Q; (D)- P

18) A particle is projected horizontally with a speed of 40m/s, from some height at t = 0. At what time will its velocity make 37° angle with the initial velocity 

(1) 1 sec

(2) 3 sec

(3) 4 sec

(4) 2 sec

19) Velocity of particle starting from rest varies with position according to equation v = $\sqrt{\alpha x}$. What is distance travelled by particle in t second from start :-

(1) $\frac{1}{2}\alpha t^2$

(2) $\frac{1}{4}\alpha t^2$

(3) $\frac{1}{3}\alpha t^3$

(4) $\frac{1}{6}\alpha t^3$

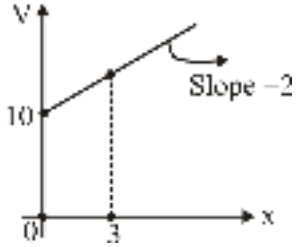
20) A particle is thrown with a speed of 12 m/s at an angle 60° with the horizontal. The time interval between the moments when its speed is 10 m/s is (g = 10 m/s^2)

(1) 1.0 s

(2) 1.2 s

(3) 1.4 s

(4) 1.6 s

21) A particle is moving along a straight line whose velocity position graph is as shown in figure. The acceleration of particle when it is at x = 3m is:- 

(1) 20 m/s^2

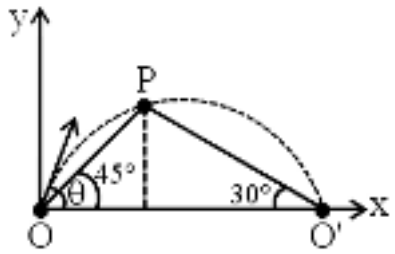
(2) 32 m/s^2

(3) 16 m/s^2

(4) 40 m/s^2

22)

For a projectile, angle of elevations from O and O' are 45° and 30° respectively at position P. Angle of projection (θ) will be :



(1) $\tan^{-1}\left(\frac{\sqrt{3}+1}{\sqrt{3}}\right)$

(2) $\sin^{-1}\left(\frac{\sqrt{3}+1}{\sqrt{3}}\right)$

(3) $\tan^{-1}\left(\frac{\sqrt{3+1}}{\sqrt{3}}\right)$

(4) $\sin^{-1}\left(\frac{\sqrt{3+1}}{\sqrt{3}}\right)$

23) A person aiming to reach the exact opposite point on the bank of stream is swimming with a speed of 0.5 m/s relative to water at angle of 120° with the direction of flow of water. Find out the speed of water :

(1) 1 m/s

(2) 0.5m/s

(3) 0.25 m/s

(4) 0.43 m/s

24) A ball is thrown vertically upwards with a velocity of 25 ms^{-1} from the top of a tower of height 30 m. How long will it travel before it hits ground?

(1) 6 s

(2) 5 s

(3) 4 s

(4) 12 s

25) A particle is projected from ground with speed 80 m/sec at an angle 30° with horizontal from ground. The magnitude of average velocity of particle in time interval t = 2 sec to t = 6 sec in (If g = 10 m/sec^2) :-

(1) $40\sqrt{2}\,m/sec$

(2) 40 m/sec

(3) zero

(4) $40\sqrt{3}\,m/sec$

26) A monkey of mass 40 kg climbs on a massless rope of breaking strength 600 N. The rope will break if the monkey :-

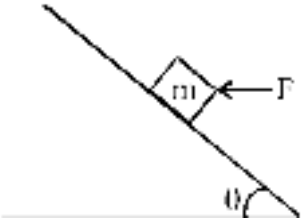
(1) Climbs up with a uniform speed of 5 m/s

(2) Climbs up with an acceleration of 5 m/s^2

(3) Climbs down with an acceleration of 4 m/s^2

(4) Climbs down with a uniform speed of 5 m/s

27) If block is in equilibrium on a smooth inclined plane then value of normal reaction by incline is:-



(1) mg cosθ

(2) mg sinθ

(3) mg/cosθ

(4) mg/sinθ

28) A block of mass 4kg is placed on a rough horizontal plane. A time dependent force $F = kt^2$ act on the block, Where $k = 2N/s^2$. Coefficient of friction μ is 0.8. Force of friction between block and the plane at t = 2 sec is :-

(1) 8 N

(2) 4 N

(3) 2 N

(4) 32 N

29) The pendulum of mass 1 kg hanging from the ceiling of a railway carriage make an angle θ with the vertical, when it is accelerating with $\frac{g}{\sqrt{3}} m/s^2$. The tension in the string of the pendulum is :-

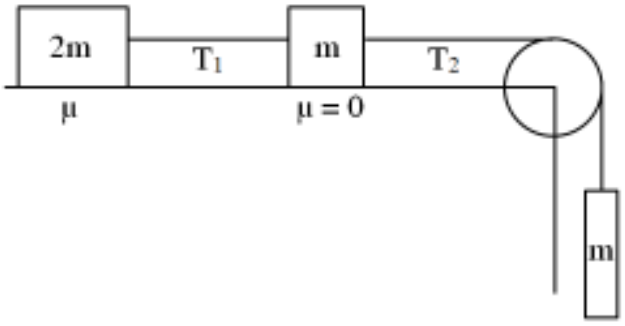
(1) $\frac{\sqrt{3}}{2}N$

(2) $\frac{2g}{\sqrt{3}}N$

(3) $g\sqrt{3}N$

(4) $\frac{g}{\sqrt{3}}N$

30) Three blocks of masses 2m, m and m are connected with two massless inextensible strings as



shown in figure : Friction exists only between block of mass 2m and table and has coefficient μ. Minimum value of μ so that system remains stationary is :

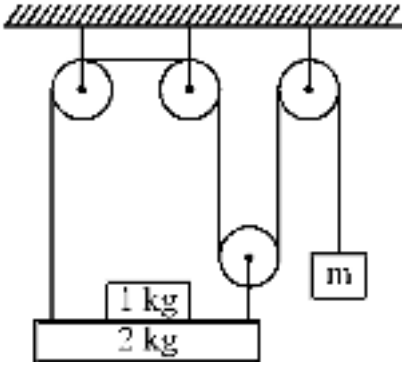
(1) $\frac{1}{4}$

(2) $\frac{1}{3}$

(3) $\frac{1}{2}$

(4) 1

31) If the arrangement shown in the figure is in equilibrium then mass m of the block will be :-



(1) 3 kg

(2) 2 kg

(3) 1.5 kg

(4) 1 kg

32) If the blocks moves with same acceleration then change in the length of spring is :



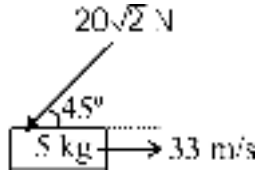
(1) 0.52 m

(2) 0.2 m

(3) 0.1 m

(4) Zero

33) A block of mass 5 kg is kept on a rough horizontal floor. It is given a velocity 33 m/s towards right. A force of $20\sqrt{2}\,\text{N}$ continuously acts on the block as shown in the figure. If the coefficient of friction between block and floor is 0.5, the velocity of block after 3 seconds is (g = 10 m/s^2) :-



(1) 2 m/s

(2) Zero

(3) 33/12 m/s

(4) None of above

34) A body is sliding down a rough inclined plane. The coefficient of friction between the body and the plane is 0.5. The ratio of the net force required for the body to slide down and the normal reaction on the body is 1:2. Then the angle of the inclined plane is:-

(1) 15°

(2) 30°

(3) 45°

(4) 60°

35) A machine gun is mounted on a 2000 kg vehicle on a horizontal smooth road (friction negligible). The gun fires 10 bullets per sec with a velocity of 500 m/s. If the mass of each bullet be 10 g, what is the acceleration produced in the vehicle ?

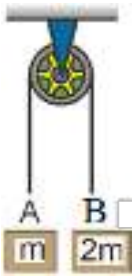
(1) 2.5 cm/s^2

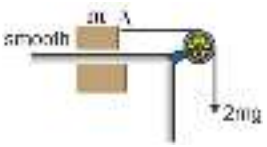
(2) 25 cm/s^2

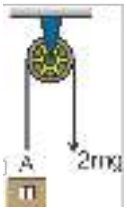
(3) 50 cm/s^2

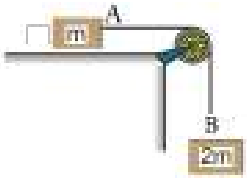
(4) 25 m/s^2

36) In which of the following cases the magnitude of acceleration of the block A will be maximum (Neglect friction, mass of pulley and string) ?

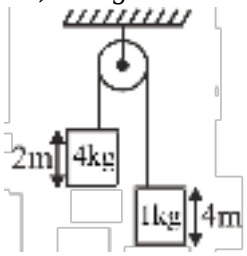
(1)


(2)


(3)


(4) 

37) In figure shown, both blocks are released from rest. Find the time to cross each other ?



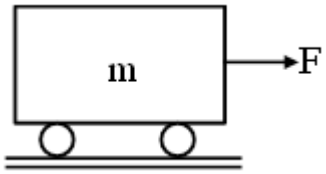
(1) 1 sec

(2) $\sqrt{2}$ sec

(3) 2 sec

(4) $\frac{1}{\sqrt{2}}$ sec

38) A car of mass m is accelerating on a level smooth road under the action of a single force F. The power delivered to the car is constant and equal to P. If the velocity of the car at an instant is v, then after travelling how much distance its speed becomes double ?



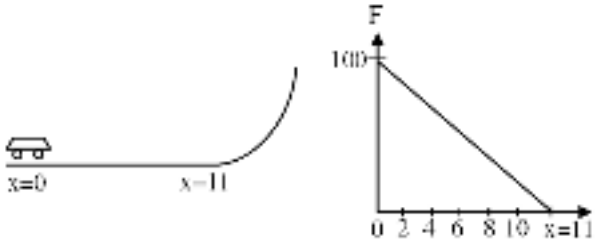
(1) $\frac{7mv^3}{3P}$

(2) $\frac{4mv^3}{3P}$

(3) $\frac{mv^3}{P}$

(4) $\frac{18mv^3}{7P}$

39) A toy car of mass 5 kg moves up a ramp under the influence of force F plotted against displacement x. The maximum height attained is given by :-



(1) h_{max} = 20 m

(2) h_{max} = 15 m

(3) $h_{max} = 11$ m

(4) $h_{max} = 5$ m

40) A spring of spring constant k placed horizontally on a rough horizontal surface is compressed against a block of mass m placed on the surface so as to store maximum energy in the spring. After removing the external force spring and block system is at rest. If the coefficient of friction between the block and the surface is μ, the potential energy stored in the spring is :

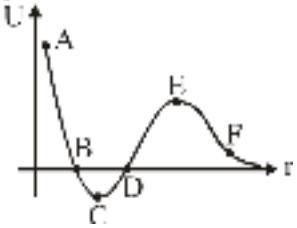
(1) $\frac{\mu^2 m^2 g^2}{k}$

(2) $\frac{2\mu m^2 g^2}{k}$

(3) $\frac{\mu^2 m^2 g^2}{2k}$

(4) $\frac{3\mu^2 mg^2}{k}$

41) The given plot shows the variation of U, the potential energy of interaction between two particles with the distance separating them, r. Then which of the following statements is correct?



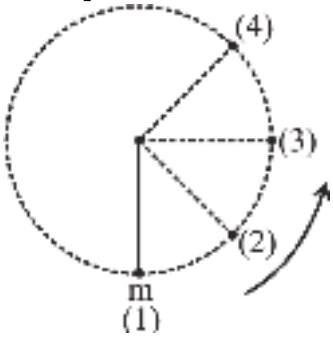
(1) B and D are equilibrium points

(2) C is point of unstable equilibrium

(3) The force of interaction between the two particles is attractive between points C and D and repulsive between points D and E on the curve.

(4) The force of interaction between the particles is repulsive between points E and F on the curve.

42) A particle (bob) of mass 1 kg is performing vertical circular motion. Then :



(1) Power delivered by tension at point (4) is negative.

(2) Power delivered by tension at point (2) is positive.

(3) Power delivered by gravitational force at point (3) is zero.

(4) Power delivered by gravitational force at point (4) is negative.

43) A particle moves in a straight line with its retardation proportional to its displacement 'x' change

in kinetic energy is proportional to :-

(1) x^2

(2) e^x

(3) x

(4) $\log_e x$

44) Consider a drop of rain water having mass 1 g falling from a height of 1 km. It hits the ground with a speed of 50 m/s. Take 'g' constant with a value 10 m/s^2. The work done by the (i) gravitational force and the (ii) resistive force of air is :

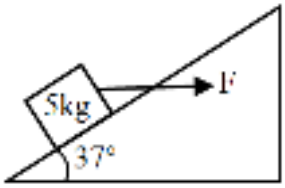
(1) (i) 1.25 J (ii) – 8.25 J

(2) (i) 100 J (ii) 8.75 J

(3) (i) 10 J (ii) – 8.75 J

(4) (i) –10 J (ii) – 8.25 J

45) A block of mass 5 kg is pulled up an inclined plane with constant velocity by applying a force as shown in figure. If the block gets displaced by 2 m, the work done by force F is :-



(1) 60 J

(2) 30 J

(3) 90 J

(4) 15 J

CHEMISTRY

1) Number of moles of methane required to produce 11 g $CO_2(g)$ after complete combustion is :-

(1) 0.35

(2) 0.5

(3) 0.75

(4) 0.25

2) Match list-I with list-II :

	List-I		**List-II**
(A)	16 g of $CH_4(g)$	(I)	weighs 28 g
(B)	1 g of $H_2(g)$	(II)	has 4 mole H atoms

(C)	1 mol of N_2 (g)	(III)	occupies 11.35 L at STP
(D)	0.5 mol of SO_2 (g)	(IV)	weighs 32 g

Choose the correct answer from the options given below :

(1) A-III, B-I, C-II, D-IV

(2) A-II, B-III, C-I, D-IV

(3) A-I, B-III, C-II, D-IV

(4) A-II, B-III, C-IV, D-I

3) If 5 moles of $BaCl_2$(aq.) is mixed with 2 moles of Na_3PO_4(aq.), the maximum number of moles of $Ba_3(PO_4)_2$ formed is :-

(1) 1.66

(2) 1

(3) 10

(4) 5

4) Haemoglobin contains 0.34% iron by mass. The number of Fe atoms in 2.8 g of Haemoglobin is :-
(Atomic mass of Fe = 56 u)
($N_A = 6 \times 10^{23}$)

(1) 6.8×10^{18}

(2) 6.8×10^{16}

(3) 1.02×10^{20}

(4) 1.02×10^{18}

5) Among the following statements, which was not proposed by Dalton's theory.

(1) Chemical reaction involve reorganisation of atoms, these are neither created nor destroyed in chemical reaction.

(2) All the atoms of a given elements have identical properties including identical mass.

(3) Matter consist of indivisible atoms.

(4) When gases combine or reproduced in a chemical reaction, they do so in a simple ratio by volume provided all gases are at same T & P.

6) Mass of ammonia produced when 2.8 kg dinitrogen quantitatively reacts with 2 kg of dihydrogen is :-

(1) 1.13 kg

(2) 3.4 kg

(3) 4.8 kg

(4) 2.8 kg

7) Maximum and minimum number of atoms are in respectively :-

(a) 16 g O_3 (b) 28 g CO
(c) 4.4 g CO_2 (d) 44 g N_2O

(1) a, d

(2) b, d

(3) d, c

(4) b, c

8) Following reaction occurs in blast furnace
$Fe_2O_3(s) + 3CO(g) \rightleftharpoons 2Fe(\square) + 3CO_2(g)$
Which of the following will not disturb the equilibrium

(1) Addition of CO_2

(2) Removal of CO_2

(3) Removal of CO

(4) Addition of Fe_2O_3

9) Equilibrium constant for the reaction
$SO_3(g) \rightleftharpoons SO_2(g) + \frac{1}{2}O_2(g)$ is 4.9×10^{-2} then equilibrium constant value for reaction
$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ is :-

(1) 4.9

(2) 49

(3) 41.6

(4) 416

10) In which case equilibrium mixture has negligible concentration of product(s) :-

(1) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
$K_C = 6.8 \times 10^5$

(2) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$
$K_C = 3 \times 10^4$

(3) $N_2O_4(g) \rightleftharpoons 2NO_2(g)$
$K_C = 1700$

(4) $2A(g) \rightleftharpoons B(g) + C(g)$
$K_C = 2 \times 10^{-3}$

11) 4.5 moles each of hydrogen (H_2) and iodine (I_2) is heated in a sealed 10 L vessel. At equilibrium 3 moles of HI were found. The equilibrium constant for
$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

(1) 1

(2) 9

(3) 4

(4) $\frac{1}{9}$

12) For the reaction

$$Fe_2N(s) + \frac{3}{2}H_2(g) \rightleftharpoons 2Fe(s) + NH_3(g)$$

(1) $K_C = K_P (RT)$

(2) $K_C = K_P (RT)^{-1/2}$

(3) $K_C = K_P (RT)^{1/2}$

(4) $K_C = K_P (RT)^{3/2}$

13) For dissociation reaction $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ correct expression of degree of dissociation (α) with K_P and total pressure at equilibrium (P) is - (assume α to be very small compared to 1)

(1) $\alpha = \sqrt{\frac{4K_P}{P}}$

(2) $\alpha = \sqrt{\frac{K_P}{2P}}$

(3) $\alpha = \sqrt{\frac{K_P}{P}}$

(4) $\alpha = \sqrt{\frac{2K_P}{P}}$

14) Equilibrium constant for the reaction A + B ⇌ C + D is 9, starting with an equimolar concentration of A, B, C and D all equal to 1M. The equimolar concentration of C is :-

(1) 0.5 M

(2) 1.5 M

(3) 0.75 M

(4) 1.25 M

15) Given below are two statements :-
Statement-I :- Aqueous solution of ammonium chloride is basic.
Statement-II :- In case of ammonium chloride, ammonium ion undergo hydrolysis to form NH_4OH and H^+ ions.
In light of above statements, choose the most appropriate answer from the options given below :-

(1) Both statement-I and statement-II are correct.

(2) Statement-I is correct but statement-II is incorrect.

(3) Both statement-I and statement-II are incorrect.

(4) Statement-I is incorrect but statement-II is correct

16) If the pK_a of weak acid HA is 5, then the pH of 0.01M of NaA(aq.) solution is :-

(1) 5.5

(2) 3.5

(3) 8.5

(4) 11.5

17) Solubility of AgCl is maximum in :-

(1) 0.1 M $AgNO_3$(aq.)

(2) 0.1 M KCl(aq.)

(3) Pure water

(4) 0.1 M NaCl(aq.)

18) For the reaction
$2A(g) + B(g) \rightarrow 2D(g)$
Change in internal energy at 298 K is –10.5 kJ, then change in enthalpy (ΔH) is :-

(1) 12.98 kJ

(2) –12.98 kJ

(3) –8.02 kJ

(4) 8.02 kJ

19) Enthalpies of formation of $CCl_4(g)$, $H_2O(g)$, $CO_2(g)$ and HCl(g) are –105, –242, –394 and –92 kJ mol^{-1} respectively, then enthalpy of given reaction is :-
$CCl_4(g) + 2H_2O(g) \rightarrow CO_2(g) + 4HCl(g)$

(1) +173 kJ

(2) –1351 kJ

(3) –173 kJ

(4) +1351 kJ

20) Among the following select the set of intensive properties :-

(1) Volume, temperature, moles

(2) Viscosity, pressure, boiling point

(3) Entropy, temperature, density

(4) Mass, volume, moles

21) The enthalpy of vapourisation of a liquid is 30 kJ mol^{-1} and entropy of vapourisation is 75 J $mol^{-1}K^{-1}$, then boiling point of liquid at 1 atm is :-

(1) 250 K

(2) 400 K

(3) 450 K

(4) 600 K

22) Two reactions are given below
$2Fe(s) + \frac{3}{2}O_2(g) \rightarrow Fe_2O_3(s)$; ΔH° = –822 kJ/mol
$C(s) + \frac{1}{2}O_2(g) \rightarrow CO(g)$; ΔH° = –110 kJ/mol
then enthalpy change of reaction

$3C(s) + Fe_2O_3(s) \rightarrow 2Fe(s) + 3CO(g)$, is :-

(1) 492 kJ

(2) –492 kJ

(3) –1152 kJ

(4) +1152 kJ

23) An ideal gas expand in volume from 1 L to 10 L at 300 K against a constant pressure of 1 atm, then work done is :- (Use 1 atm-L = 100 J)

(1) –900 J

(2) –900 kJ

(3) 2710 kJ

(4) 900 kJ

24) Solubility product (K_{sp}) of zirconium phosphate $Zr_3(PO_4)_4$ in terms of its molar solubility(s) is given by :-

(1) $6912\ s^7$

(2) $5348\ s^6$

(3) $8435\ s^7$

(4) $9612\ s^3$

25) Which of the following is a redox reaction ?

(1) $2Na\ [Ag(CN)_2] + Zn \rightarrow Na_2\ [Zn(CN)_4] + 2Ag$

(2) $BaO_2 + H_2SO_4 \rightarrow BaSO_4 + H_2O_2$

(3) $N_2O_5 + H_2O \rightarrow 2HNO_3$

(4) $AgNO_3 + KI \rightarrow AgI + KNO_3$

26) Concentration of X^- ion in an aqueous solution containing 0.1 M HCl and 0.2 M HX
$\{K_a(HX) = 10^{-5}\}$ $\{\log 2 = 0.3\}$

(1) 0.2 M

(2) 10^{-5} M

(3) 2×10^{-5} M

(4) 3×10^{-5} M

27) 80 mL of a gaseous hydrocarbon is mixed with 500 mL O_2 and ignited. After combustion a gaseous mixture of 340 mL is collected at NTP, which upon passing through KOH solution, results in 100 mL of gas remaining then, hydrocarbon is :

(1) C_2H_4

(2) C_3H_6

(3) C_3H_8

(4) C_4H_{10}

28) Processes in which entropy of system increases :-
(A) $H_2(g) \rightarrow 2H(g)$
(B) Evaporation of liquid
(C) Isothermal compression of ideal gas from 20 L to 10 L at 300 K.
(D) $2NaHCO_3(s) \rightarrow Na_2CO_3(s) + H_2O(g) + CO_2(g)$
(E) Mixing of two gases
Select the correct option :-

(1) A, C and D

(2) A, B, D and E

(3) Only C

(4) A, B and D

29) For reaction $NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$ at 25°C, equilibrium pressure is 3 atm then K_p of reaction is :-

(1) 2.25

(2) 9

(3) 1

(4) 0.45

30) Oxidation number of oxygen in H_2O_2 and KO_2 respectively is :-

(1) $-\frac{1}{2}$ and –1

(2) $-\frac{1}{2}$ and +1

(3) –1 and +2

(4) –1 and +1

31) Match list-I with list-II :-

	List-I Reaction		**List-II Type of Redox reaction**
(A)	$2KClO_3(s) \rightarrow 2KCl(s) + 3O_2(g)$	(P)	Combination reaction
(B)	$Cr_2O_3(s) + 2Al(s) \rightarrow Al_2O_3(s) + 2Cr(s)$	(Q)	Disproportionation reaction
(C)	$2H_2O_2(aq.) \rightarrow 2H_2O(\square) + O_2(g)$	(R)	Decomposition reaction
(D)	$C(s) + O_2(g) \rightarrow CO_2(g)$	(S)	Displacement reaction

Select the correct option :-

(1) A-P, B-Q, C-R, D-S

(2) A-Q, B-R, C-S, D-P

(3) A-R, B-S, C-Q, D-P

(4) A-R, B-Q, C-S, D-P

32) $2IO_3^- + xI^- + 12H^+ \rightarrow 6I_2 + 6H_2O$

value of x is :-

(1) 2

(2) 12

(3) 10

(4) 6

33) 20mL of 0.2M $K_2Cr_2O_7$ solution is used for the titration of 10 mL Fe^{2+} solution in acidic medium. Molarity of Fe^{2+} solution is :-

(1) 1.2 M

(2) 0.8 M

(3) 2.4 M

(4) 1.6 M

34) Solubility product of $BaSO_4(s)$ is 1×10^{-10} at 298 K, then solubility product of $BaSO_4(s)$ in 0.1 M $K_2SO_4(aq.)$ solution at 298 K is :-

(1) 1×10^{-9}

(2) 1×10^{-10}

(3) 1×10^{-8}

(4) 1×10^{-5}

35) 400 mL of 0.01 M HCl (aq.) solution is mixed with 600 mL of 0.01 M HNO_3 (aq.) solution, then pH of resultant solution is :-

(1) 4

(2) 2

(3) 1.7

(4) 2.3

36) Correct increasing order of pH of 0.1 M aq. solution of each of HCl, NaOH, NH_4Cl and NaCl is :-

(1) NaOH < NH_4Cl < NaCl < HCl

(2) HCl < NH_4Cl < NaCl < NaOH

(3) HCl < NaCl < NH_4Cl < NaOH

(4) HCl < NH_4Cl < NaOH < NaCl

37) Two sparingly soluble salts AX_2 and MX have same value of solubility product 4×10^{-12}, then ratio of their molar solubilities is :-

$\frac{s(AX_2)}{s(MX)} =$

(1) 1 : 50

(2) 50 : 1

(3) 1 : 100

(4) 2 : 15

38) Given below are two statements :
Statement-I :- pH of water increases with increase in temperature.
Statement-II :- Dissociation of water in H^+ and OH^- ion is an endothermic process.
In light of above statements, select the correct option.

(1) Both statements are correct.

(2) Both statements are incorrect.

(3) Statement-I is correct but statement-II is incorrect.

(4) Statement-I is incorrect but statement-II is correct.

39) Select the correct statement(s) for 0.01 M NH_4Cl (aq.) solution ($K_b = 2 \times 10^{-5}$)
(A) Cationic hydrolysis take place.
(B) Hydrolysis constant (K_h) is 5×10^{-10}.
(C) pH of solution is 5.65.
(D) Nature of solution is basic.

(1) A, B and C

(2) A, C and D

(3) B, C and D

(4) A, B and D

40) For a reversible reaction at equilibrium A ⇌ B, rate constants of forward and backward reactions are respectively 10^3 and 10^2, then standard Gibbs energy change ($\Delta_r G°$) of reaction at 27°C is :-

(1) –5.75 kJ

(2) 5.75 kJ

(3) –0.46 kJ

(4) 0.46 kJ

41) For a given reaction A → B, at 300 K, the free energy change is –50kJ/mol and heat of reaction is 51.4 kJ/mol then entropy change of reaction is :-

(1) 4.66 J mol^{-1} K^{-1}

(2) 33.8 J mol^{-1} K^{-1}

(3) 338 J mol^{-1} K^{-1}

(4) 3.38 J mol^{-1} K^{-1}

42) When 2L of an ideal gas expands isothermally into vacuum to a total volume of 6L, the change in internal energy is :- (log3 = 0.48)

(1) 0

(2) –2.7 kJ

(3) +2.7 kJ

(4) –0.4 kJ

43) Match list-I with list-II :

	List-I		**List-II**
(A)	$\Delta H = - ; \Delta S = +$	(i)	Reaction is non-spontaneous at all temperatures
(B)	$\Delta H = + ; \Delta S = -$	(ii)	Reaction is spontaneous at high temperature
(C)	$\Delta H = + ; \Delta S = +$	(iii)	Reaction is spontaneous at all temperature
(D)	$\Delta H = - ; \Delta S = -$	(iv)	Reaction is spontaneous at low temperature

(1) A-(ii), B-(iii), C-(i), D-(iv)

(2) A-(iii), B-(i), C-(ii), D-(iv)

(3) A-(i), B-(iii), C-(ii), D-(iv)

(4) A-(iv), B-(iii), C-(i), D-(ii)

44) From following, select the option with state variable(s) :-
(A) Pressure (P)
(B) Enthalpy (H)
(C) Work (W)
(D) Entropy (S)
(E) Temperature (T)

(1) A, B and E

(2) A, B, D and E

(3) C, D and E

(4) B, D and E

45) Heat of atomisation of $CH_4(g)$ and $C_2H_6(g)$ are x kJ mol^{-1} and y kJ mol^{-1} then BE(C–C) is :-

(1) (x + y) kJ mol^{-1}

(2) (y – x) kJ mol^{-1}

(3) $\left(y - \frac{2x}{3}\right) kJ\,mol^{-1}$

(4) $\left(y-\frac{3x}{2}\right)$ kJ mol^{-1}

BIOLOGY

1) Read the following statements and select the correct ones :-
(A) Increase in mass and increase in number of individuals are twin characters of growth.
(B) Response to stimuli is the defining properpty of living organisms.
(C) Metabolic reactions can be demonstrated outside the body in isolated cell free system.
(D) All organisms are not aware of their surroundings.

(1) Only B and C
(2) Only A and C
(3) A, B and C
(4) A, B and D

2) Which of the following is NOT a correct statement?

(1) Euglenoids have a protein rich layer instead of cell wall
(2) Slime moulds are decomposer protists
(3) Dinoflagellates cell wall is indestructible
(4) Diatoms are cheif producers of oceans

3) Identify the correct match from **column I, II** and **III**.

Column I (Kingdom)		**Column II (Cell wall)**		**Column III (Body organisation)**	
1	Monera	a	Present in some	i	Tissue/ organ
2	Protista	b	Present (Chitin)	ii	Loose tissue
3	Fungi	c	Present (Cellulose)	iii	Cellular
4	Plantae	d	Polysaccharide + Amino acid	iv	Cellular

(1) 1–d–iv, 2–a–iii, 3–b–ii, 4–c–i
(2) 1–d–iv, 2–b–i, 3–a–ii, 4–c–iii
(3) 1–d–iv, 2–c–ii, 3–a–i, 4–b–iii
(4) 1–c–iv, 2–b–iii, 3–d–ii, 4–c–i

4) Single protective unit of bacteria is made up of:-

(1) Cell wall and cell membrane only
(2) Glycocalyx, cell wall and flagella
(3) Cell wall, cell membrane and pili
(4) Capsule, cell wall and cell membrane

5) Bacterial structure and behaviour are respectively:

(1) Simple, Simple
(2) Complex, Simple
(3) Simple, Complex
(4) Complex, Complex

6) Choose the incorrect statement from following :-

(1) Ribosomes are the site of protein synthesis
(2) Pili do not play a role in motility
(3) Inclusion bodies are bounded by membrane
(4) Cell wall prevent the bacterium from bursting and collapsing

7) Which of the following is incorrect about blue green algae ?

(1) They may be unicellular, colonial or filamentous
(2) Their colonies are generally surrounded with membranous sheath
(3) They often form blooms in polluted water bodies
(4) Some of the members can fix atmospheric nitrogen

8) Which set of disease is caused by bacteria?

(1) Cholera, AIDS, Tetanus
(2) Cholera, Typhoid, Tetanus
(3) Typhoid, Tetanus, Influenza
(4) AIDS, Mumps, Kuru

9) Choose the incorrect statement regarding bacteria :-

(1) They are sole member of kingdom monera
(2) They show most extensive metabolic diversity
(3) They play role in ecology as a decomposer
(4) They show advance type of cellular organization

10) Match the items of **column-I** with **column-II** and select correct option.

Column-I		**Column-II**	
(a)	Mesosome	(i)	Extra genomic DNA
(b)	Plasmid	(ii)	Helps in nitrogen fixation
(c)	Chromatophores	(iii)	Photosynthetic structure in some prokaryotes
(d)	Heterocyst	(iv)	Cell wall formation

(1) a-(iv), b-(i), c-(iii), d-(ii)
(2) a-(i), b-(ii), c-(iii), d-(iv)
(3) a-(iii), b-(i), c-(iv), d-(ii)

(4) a-(iii), b-(iv), c-(i), d-(ii)

11)

Read the following terms carefully.
Cilia, Flagella, Zygote, Diatomite, Photosynthesis, Pellicle, Peptidoglycan wall, Bioluminescence, Multicellular, Heterocyst, Decomposer, Pseudopodia.

How many terms are not related with kingdom Protista:-

(1) 4

(2) 5

(3) 3

(4) 6

12) Which of the following is not concerned with protista kingdom ?

(1) Eukaryotic Organisation.

(2) Unicellular Organism

(3) Asexual and Sexual reproduction

(4) Boundaries of this kingdom is well defined

13) Find the correct option for the protista kingdom, referring to the organisms present in the given box, which are photosynthetic with cell walls and photosynthetic without cell walls respectively ?

Diatoms, Desmids, *Gonyaulax* , *Euglena*, slime moulds, *Amoeba*, *Entamoeba*, *Fucus*, *Paramoecium*, *Plasmodium*

(1) 4 and 1

(2) 3 and 3

(3) 3 and 1

(4) 3 and 7

14) **Assertion (A) :-** In pteridophytes, the female gametophytes are retained on the parent sporophyte for variable periods.
Reason (R) :- In these plants the development of the zygote into young embryo take place with in the female gametophyte.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

15) **Assertion (A) :-** Bryophytes are called amphibians of the plant kingdom.
Reason (R) :- They play an important role in plant succession.

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(3) **Assertion** is true but **Reason** is false.

(4) **Assertion** is false but **Reason** is true.

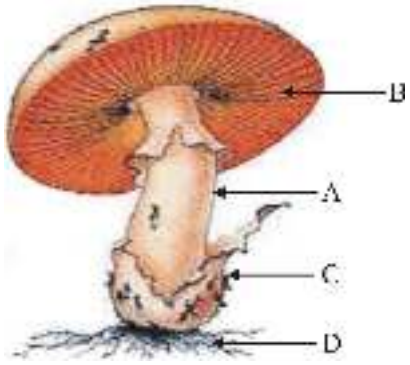
16) Read the following statements & select the correct option.
Statement-I :- Groups represent category, category further denotes rank and each rank or taxon represents a unit of classification.
Statement-II :- The taxonomic categories are not distinct biological entities and are merely morphological aggregates.

(1) Both statement-I and statement-II are correct.
(2) Both statement-I and statement-II are incorrect.
(3) Statement-I is correct but statement-II is incorrect.
(4) Statement-I is incorrect but statement-II is correct.

17) Which of the following belongs to phycomycetes ?

(1) *Alternaria, Rhizopus*
(2) *Albugo, Rhizopus*
(3) Mushroom, *Claviceps*
(4) *Penicillium*, Yeast

18) According to given diagram select the part that produces basidiospore :-



(1) A
(2) B
(3) C
(4) D

19)

Deuteromycetes are commonly called 'fungi imperfecti' because:

(1) Only sexual phases of these fungi are known
(2) Both sexual and asexual phases are known
(3) Only asexual or vegetative phases are known
(4) They possess fruiting bodies

20) Which one of the following is mismatched :-

(1) *Penicillium* - Source of antibiotics

(2) *Albugo* - Parasitic fungi on legume

(3) *Neurospora* - Used in biochemical and genetic work

(4) *Agaricus* - Lack sex organ

21) Who demonstrated that the extract of the infected plants of tobacco could cause infection in healthy plants ?

(1) Ivanowsky

(2) Beijerinck

(3) Stanley

(4) Pasteur

22) Match the **column-I, column-II** and **column-III** and select the correct option from option given below :-

Column I		**Column II**		**Column III**	
A	Virus	P	Low molecular weight RNA	i	Potato spindle tuber disease
B	Lichen	Q	Folded protein	ii	Indicator of air pollution
C	Prions	R	Helotism	iii	Cr-Jacob disease
D	Viroids	S	Protein coat present	iv	T.M.V.

(1) A-S-iv, B-R-ii, C-Q-iii, D-P-i

(2) A-S-iii, B-P-i, C-R-ii, D-Q-iv

(3) A-Q-ii, B-S-iii, C-P-iv, D-R-i

(4) A-P-i, B-Q-iv, C-R-iii, D-S-ii

23) Which of the following group of marine algae is used as food ?

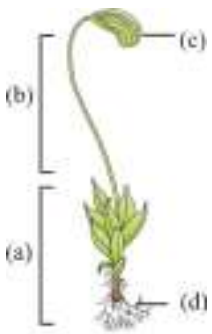
(1) *Chlamydomonas*, *Volvox* and *Gracilaria*

(2) *Porphyra, Laminaria* and *Sargassum*

(3) *Laminaria* and *Gracilaria*

(4) *Porphyra* and *Chlamydomonas*

24) Examine the figure given below and select the correct option, ensuring all the four parts (a, b, c and d) are correctly identified :-



	a	**b**	**c**	**d**
1	Gametophyte	Sporophyte	Capsule	Rhizome

2	Sporophyte	Capsule	Rhizoids	Gametophyte
3	Gametophyte	Capsule	Sporophyte	Rhizoids
4	Gametophyte	Sporophyte	Capsule	Rhizoids

(1) 1

(2) 2

(3) 3

(4) 4

25) Read the following statements (i-iii) and answer the question given below -
(i) Vegetative reproduction usually take place by fragmentation.
(ii) Asexual reproduction is by 2-8 flagellated zoospores.
(iii) Pyrenoids contain protein besides starch.
Which option is related to the given statements (i-iii) ?

(1) Diatom

(2) Chlorophyceae

(3) Phaeophyceae

(4) Rhodophyceae

26) What is common among *Morchella*, *Agaricus*, *Chlorella* and *Spirulina* ?

(1) They all are placed under same kingdom according to five kingdom classification.

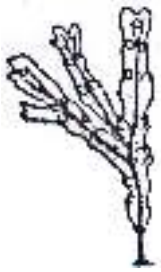
(2) They all have nuclear membrane

(3) They all may be used as food or food supplement.

(4) They all are autotrophic

27) Select the incorrect match :

(1)		Green algae, presence of pyrenoids, starch is the reserve food
(2)		Red algae, presence of phycoerythrin pigment

(3)		Green algae, Fucoxanthin pigment, cell wall with polysulphate esters
(4)		Brown algae, two flagella in zoospore

(1) 1

(2) 2

(3) 3

(4) 4

28) Consider the following characteristics.
(a) Occurence of true roots, stem and leaves
(b) Dependent sporophyte
(c) Presence of vascular tissue
(d) Independent, free-living gametophytes
(e) Presence of archegonia
(f) Formation of seeds
How many of the above are similar between pteridophytes and gymnosperms ?

(1) Two

(2) Four

(3) Five

(4) Three

29) Read the following **statements** carefully and select the correct answer :-
(A) Both Phaeophyceae and Rhodophyceae are primarily found in marine habitats.
(B) Gemmae are green and multicellular buds, produced on ventral surface of thallus for sexual reproduction.
(C) In pteridophytes, spores germinate to give rise to inconspicuous, small but multicellular and free living nonvascular gametophyte.
(D) In gymnosperms male gametes are transferred through pollen tube and the process is called as siphonogamy.
Which of the following options is most correct from the following:

(1) A, B - correct & C, D - incorrect

(2) C, D - correct & A, B - incorrect

(3) A, C, D - correct & B - incorrect

(4) A, B, C - correct & D - incorrect

30) Some characters/structures are given below. How many of them are found in both bryophyta and

algae ?
(A) Archegonium
(B) Embryo
(C) Motile male gametes
(D) Oogamy
(E) Vascular tissue
(F) Antheridium

(1) Two
(2) Three
(3) Four
(4) Five

31) Read the following statements :
(A) More than one archegonia
(B) Show symbiotic association with nitrogen fixing cyanobacteria and fungus
(C) Show heterosporous nature
(D) Reduced male gametophyte is called a pollen grain
Above statements are related to which group of plants ?

(1) Bryophytes
(2) Gymnosperms
(3) Pteridophytes and Gymnosperm
(4) Gymnosperm and Angiosperm

32) Choose the incorrect match from following :-

(1) *Cycas* - branched stem
(2) *Salvinia* - heterosporous sporophyte
(3) *Marchantia* - dioecious
(4) *Chlorella* - used by space travellers.

33) The spread of living bryophytes and pteridophytes is restricted to moist, damp and shady places because :-

(1) They lack roots and vascular tissue
(2) Not able to tolerate high light intensity
(3) They require water for fertilization
(4) Sporophytic generation depends upon gametophytic generation for water

34) Which of the following have multicellular sporophyte which is not free living but attached to photosynthetic gametophyte and derives nourishment from it ?

(1) Pteridophytes
(2) Gymnosperms
(3) Bryophytes
(4) Thallophytes

35) Which haploid structure is formed in the life cycle of bryophytes between zygote and main plant body ?

(1) Prothallus

(2) Spore mother cell

(3) Spore

(4) Embryo

36) Read the following paragraph and fill in the blanks with appropriate words :-
In pteridophytes the main plant body is ___**(A)**____ which bears sporangia that are subtended by leaf-like appendages called ___**(B)**___. The sporangia produce spores by ___**(C)**___ and after germination of spores, they give rise to photosynthetic gametophytes called ___**(D)**___.

	A	**B**	**C**	**D**
(1)	Sporophyte	Sporangia	Meiosis	Protonema
(2)	Gametophyte	Sporophylls	Mitosis	Sporophyte
(3)	Sporophyte	Sporophylls	Meiosis	Prothallus
(4)	Gametophyte	Sporophylls	Mitosis	Protonema

(1) 1

(2) 2

(3) 3

(4) 4

37) Read the following events :-
(A) Spore formation (B) Gametes formation
(C) Zygote formation (D) Embryo formation
(E) Fertilization (F) Sex organ formation
Choose the correct sequence of above events regarding life cycle of pteridophytes.

(1) F, A, B, E, C, D

(2) A, F, B, E, D, C

(3) A, B, F, E, C, D

(4) A, F, B, E, C, D

38) Mycorrhizae is found in

(1) *Pinus*

(2) *Selaginella*

(3) Wheat

(4) *Fucus*

39) Choose the correct statement regarding gymnosperm -

(1) Seeds are developed before fertilization

(2) Ovules are not enclosed by any ovary wall and remain exposed, both before and after fertilization

(3) They show double fertilization

(4) They form haploid endosperm after fertilization

40) Animals, mammals, dogs all are taxa. But you know that a dog is a mammal and mammals are animals. Therefore animals, mammals and dogs represent :-

(1) Taxa at same levels

(2) Taxa / Taxon at different levels

(3) Taxa which have same morphological characters

(4) Taxa at division level

41)

Both the words in a biological name when handwritten are underlined or printed in italics to indicate :-

(1) They are endangered

(2) They are living

(3) Their Latin origin

(4) Now they are extinct

42) Which of the following is a **wrong** statement regarding the universal rules of nomenclature ?

(1) Biological names are generally in latin.

(2) Both the words of biological name when hand written must be cumulatively underlined.

(3) The second component of biological name denotes specific epithet.

(4) The specific epithet starts with a small letter.

43) Select the **incorrect** statement with respect to the taxon, 'genus' :-

(1) It is a group or assemblage of related species.

(2) A genus essentially comprises more than one number of species always.

(3) Lion, Tiger, Leopard are closely related species which have been placed in the genus *Panthera*.

(4) *Solanum, Mangifera* and *Homo* are the examples of genera.

44)

Carnivora includes:

(1) Group of organism belonging to producer in ecosystem

(2) Group of organism belonging to related species which are herbivorous only

(3) Group of organism belonging to related families

(4) Group of organism which are similar in all features can reproduce with each other

45) Study the following table which shows different organisms with their taxonomic categories :-

Common name	Family	Order	Class	Phylum/Division
Man	Hominidae	Primata	Mammalia	**A**
Housefly	Muscidae	Diptera	**B**	Arthropoda
Mango	**C**	Sapindales	Dicotyledonae	Angiospermae
Wheat	Poaceae	Poales	**D**	Angiospermae

Select the correct option for A, B, C and D.

(1) A–Chordata, B–Insecta,
C–Anacardiaceae, D–Monocotyledonae

(2) A–Animalia, B–Insecta,
C–Anacardiaceae, D–Dicotyledonae

(3) A–Chordata, B–Arachnida,
C–*Mangifera*, D–Monocotyledonae

(4) A–Non-chordata, B–Insecta,
C–Anacardiaceae, D–Dicotyledonae

46) **Assertion :-** The water vascular system of echinoderms helps in locomotion, capture and transport of food and respiration..
Reason :- Echinoderms are spiny bodied animals in which development is indirect with free-swimming larva.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true and (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

47) Match the column-I with column-II & select the correct option :-

Column-I		Column-II	
A.	Filaria worm	(i)	*Wuchereria*
B.	Cuttle fish	(ii)	*Adamsia*
C.	Sea Pen	(iii)	*Sepia*
D.	Sea anemone	(iv)	*Pennatula*

(1) A–i, B–ii, C–iii, D–iv

(2) A–i, B–iii, C–iv, D–ii

(3) A–iii, B–ii, C–i, D–iv

(4) A–ii, B–iii, C–iv, D–i

48) In which one of the following the genus name, its two character and its class are correctly matched :-

	Genus	Two character	Class

(1)	*Felis*	(a) External ears or pinnae are absent (b) Homeothermic	Mammalia
(2)	*Struthio*	(a) Can fly (b) Endoskeleton fully ossified	Aves
(3)	*Bangarus*	(a) epidermal scales (b) Poisonous snake	Reptilia
(4)	*Ichthyophis*	(a) 2 pair of limbs present (b) Poikilothermic animal	Amphibia

(1) 1

(2) 2

(3) 3

(4) 4

49) Hook worm is common name of :-

(1) *Ascaris*

(2) *Ancylostoma*

(3) *Limulus*

(4) *Fasciola*

50) Which of the following pairs of animals are similar to each other partaining to feature stated against them?

(1) *Pteropus* and *ornithorhyncus* = viviparity

(2) Garden lizard & crocodile = 4 chamber heart

(3) *Ascaris* & *Ancylostoma* = Metameric segmentation

(4) Sea horse & flying fish = Poikilothermal

51) Which one of the following statement regarding coelom is correct ?

(1) Round worms are pseudocoelomate

(2) Mollusca are acoelomate

(3) Arthropods are acoelomate

(4) Flatworms are coelomate.

52) Air bladders are present in :-

(1) Dog fish

(2) Electric fish

(3) Devil fish

(4) Clarias fish

53) In which of the following groups, all animals are monoecious?

(1) Tapeworm, *Neries, Pheretima*
(2) Leech, Tapeworm, Earthworm
(3) *Ascaris*, Tapeworm, Earthworm
(4) Star fish, *Pheretima, Sepia*

54)

Match Column-I with Column-II on basis of excretory organs.

	Column-I		**Column-II**
(a)		(i)	Malpighian tubules
(b)		(ii)	Coxal Gland
(c)		(iii)	Proboscis Gland
(d)		(iv)	Nephridia

(1) a-i, b-ii, c-iii, d-iv
(2) a-iii, b-i, c-iv, d-ii
(3) a-iv, b-iii, c-ii, d-i
(4) a-ii, b-i, c-iv, d-iii

55) Tissue level of body organisation is found in the members of which phylum ?

(1) Protozoa
(2) Porifera
(3) Coelenterata
(4) All

56) Which of the following combination is incorrect ?

(1) Nematoda - Roundworms, pseudocoelomate, ascaris

(2) Porifera - Gastrovascular cavity, coelom absent, spongilla

(3) Echinodermata - Coelom present, radially symmetrical, asterias

(4) Platyhelminthes - Triploblastic, flatworm, acoelomate

57)

Identify the fresh water bony fishes.

(1) Flying fish, Catla

(2) Magur, Sea horse

(3) Rohu, *Clarias*

(4) Saw fish, *Labeo*

58) Which of the following have cellular level of organisation ?

(1) Physalia

(2) Hydra

(3) Scypha

(4) Ctenoplana

59) **Statement-I :** All living members of the class cyclostomata are endoparasites on some fishes.
Statement-II : Cyclostomates have sucking and circular mouth with jaws.

(1) Statement-I & Statement-II both are correct.

(2) Statement-I is correct & Statement-II is incorrect.

(3) Statement-I is incorrect & Statement-II is correct.

(4) Both Statement-I & Statement-II is incorrect.

60) In coelenterates an undifferentiated layer is present in between the ectoderm and endoderm is called :

(1) Mesoderm

(2) Mesophyll

(3) Mesoglea

(4) Differentiated cellular layer

61) Find out incorrect match from the table given below :-

	Animal	**Phylum**	**Characters**
(1)	*Octopus*	Mollusca	Radula present
(2)	*Ophiura*	Echinodermata	Fertilisation is external
(3)	*Saccoglossus*	Hemichordata	Excretory organ is flame cells

(4)	*Sepia*	Mollusca	Known as cuttle fish

(1) 1

(2) 2

(3) 3

(4) 4

62) Read the following statements (A to D)–
(A) Platyhelminthes are bilaterally symmetrical, diploblastic and acoelomate animals.
(B) Cnidarian are radially symmetrical, triploblastic and acoelomate animals
(C) Porifera are radially symmetrical, triploblastic and acoelomate animals
(D) Ctenophora are bilaterally symmetrical, diploblastic and acoelomate animals
How many statements are incorrect ?

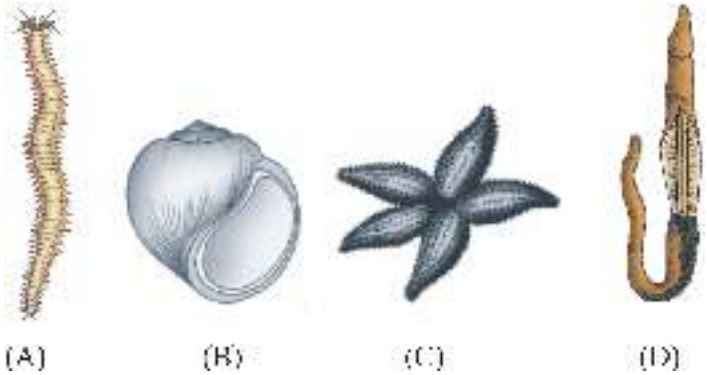
(1) One

(2) Two

(3) Three

(4) All

63) Identify the animals given in figure A,B,C and D and select the right option about them :-



(1) A = *Nereis*, C = *Balanoglossus*
B = *Pila*, D = *Asterias*

(2) A = *Balanoglossus*, C = *Pila*
B = *Asterias*, D = *Nereis*

(3) A = *Nereis*, C = *Asterias*
B = *Pila*, D = *Balanoglossus*

(4) A = *Asterias*, C = *Nereis*
B = *Balanoglossus*, D = *Pila*

64) Which of the following statement is incorrect about the given diagram :-



(1) It reproduce by sexual reproduction

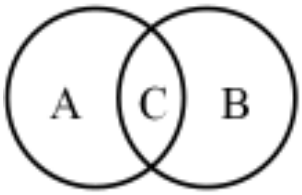
(2) Mouth and tentacles directed upward

(3) Always solitary

(4) Free swimming

65)

Observe the venn diagram shown below here :-



A = Bilateral symmetry

B = Radial symmetry

Based on the provided information, identify the phylum A, B and C :-

	A	**B**	**C**
1	Porifera	Chordata	Mollusca
2	Chordata	Echinodermata	Coelenterata
3	Platyhelminthes	Coelenterata	Echinodermata
4	Annelida	Arthropoda	Mollusca

(1) 1

(2) 2

(3) 3

(4) 4

66) Which of the following phylum is triploblastic but acoelomate?

(1) Aschelminthes

(2) Platyhelminthes

(3) Annelids

(4) Cnidarians

67) Which of the following connective tissue connects skeletal muscles to bones in form of tendon :-

(1) Areolar connective tissue

(2) Adipose tissue

(3) Dense regular connective tissue

(4) Dense irregular connective tissue

68)

Fallopian tube is lined by

(1) Ciliated cuboidal

(2) Ciliated columnar

(3) Brush bordered cuboidal

(4) Brush border columnar

69) Which of the following is not a specialised connective tissue?

(1) Blood
(2) Bone
(3) Cartilage
(4) Areolar tissue

70) Which one is the wrong statement about the "bones" ?

(1) The bone-marrow in some bones is the site of production of blood-cells.
(2) Bones have solid and pliable ground substance.
(3) Matrix of bones is rich in Ca^{+2} salts and collagen fibres.
(4) Bone cells are present in the spaces called lacunae.

71) Match the column-I with column-II and choose correct option :-

	Column-I		**Column-II**
(A)	Adipose tissue	(i)	Perform cementing to keep neighbouring cell together
(B)	Dense regular connective tissue	(ii)	Have a hard and non-pliable ground substance rich in calcium salts and collagen fibres
(C)	Bone	(iii)	Located mainly beneath the skin
(D)	Adhering junction	(iv)	Tendon and ligaments

(1) A-(iii), B-(iv), C-(i), D-(ii)
(2) A-(iv), B-(i), C-(iii), D-(ii)
(3) A-(iii), B-(ii), C-(i), D-(iv)
(4) A-(iii), B-(iv), C-(ii), D-(i)

72) Fill in the blank :-
_____ help to stop substances from leaking across a tissue :

(1) Adhering junctions
(2) Gap junctions
(3) Tight junctions
(4) All of the above

73) The sketch given below represent a type of animal tissue. 

Which of the following statement is **not** related with above tissue ?

(1) It has a limited role in secretion and absorption

(2) It's main function is to provide protection against chemical and mechanical stresses.

(3) Cells are compactly packed with little intercellular matrix.

(4) It is found in the lining of stomach and intestine.

74) The type of epithelium found in blood vessels and stomach is

(1) Stratified cuboidal in both

(2) Stratified columnar and simple cuboidal

(3) Simple squamous and simple columnar

(4) Transitional epithelium

75) For female cockroaches, how many statements are **correct**?
(a) The 7th sternum is boat shaped
(b) The 7th sternum together with the 8th and 9th sterna forms a brood or genital pouch
(c) The anterior part of genital pouch contains female gonopore, spermathecal pores and collateral glands
(d) Phallic gland form ootheca

(1) a and b

(2) b and d

(3) a, b, c

(4) a and d

76) Another name of Labrum ?

(1) Upper lip

(2) Lower lip

(3) Upper jaw

(4) Lower jaw

77) Spiracles found in cockroach are :-

(1) 8 pairs

(2) 9 pairs

(3) 10 pairs

(4) 11 pairs

78) **Assertion :-** In female cockroach, one pair of spermtheca is present which open into genital chamber.
Reason :- In female cockroach fertilised eggs are encased in dark reddish to blackish brown capsule called spermtheca.

(1) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(2) Assertion is true but Reason is false.

(3) Assertion is false but Reason is true.

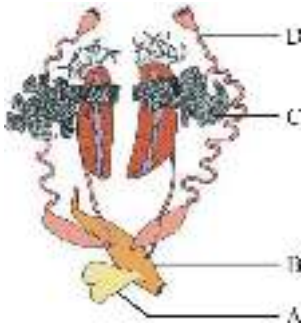
(4) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

79) Read the following statements carefully and choose the correct option.
Statement-I : Heart of cockroach consist of elongated muscular tube lying along mid ventral line of thorax and abdomen.
Statement-II : Blood from sinsuses enter heart through ostia and is pumped posteriorly to sinuses again.

(1) Both **Statement I** and **Statement II** are incorrect
(2) **Statement I** is correct but **Statement II** is incorrect
(3) **Statement I** is incorrect but **Statement II** is correct
(4) Both **Statement I** and **Statement II** are correct

80) The sclerite present on ventral side of cockroach is known as :

(1) Tergites
(2) Pleurites
(3) Sternites
(4) Tergum

81) Given below is a diagramatic representation of female reproductive system of frog labelled A, B, C and D :  Find out incorrect one :-

(1) There is functional connection between C and kidney.
(2) B is small, median chamber that is used to pass faecal matter, urine and ova to the exterior.
(3) D is one pair arising from the ovaries opens into the cloaca separately.
(4) A is thin walled urinary bladder is present ventral to rectum which also opens in cloaca.

82)

Which of the following statement is **Incorrect** with reference to frog :-
(I) Eyes are bulged and not covered by nictitating membrane.
(II) Membranous tympanum receives the sound signals.
(III) The frog never drinks water.

(IV) pair of nostrils is present above to mouth.
(1) I and II
(2) III and IV
(3) I and IV
(4) Only I

83) Adult frog is :-

(1) Uricotelic
(2) Ureotelic
(3) Ammonotelic
(4) Guanotelic

84) Select the correct route for the passage of sperms in male frogs :-

(1) Testes → Bidder's canal → Kidney → Vasa efferentia → Urinogenital duct → Cloaca.
(2) Testes → Vasa efferentia → Kidney → Bidder's canal → Urinogenital duct → Cloaca.
(3) Testes → Vasa efferentia → Kidney → Seminal vesicle → Urinogenital duct → Cloaca.
(4) Testes → Vasa efferentia → Bidder's canal → Ureter → Cloaca.

85) In frogs, testes are attached to kidneys by a double fold of peritoneum known as :-

(1) Urinary bladder
(2) Mesovarium
(3) Urinogenital duct
(4) Mesorchium

86) The digestion in frog start from ?

(1) Mouth
(2) Stomach
(3) Intestine
(4) Oesophagus

87) Which of the following mode of respiration always continue in frog ?

(1) buccopharyngeal
(2) Pulmonary
(3) Cutaneous
(4) Gills

88) Match the column-A to column-B

	Column-A		**Column-B**
(i)	Skin is uniform pale yellow	(a)	Dorsal side
(ii)	Olive green with dark irregular spots	(b)	Ventral side
(iii)	Ability to change the colour	(c)	Mimicry

(iv)	Protective colouration	(d)	Camouflage

(1) i–a, ii–b, iii–c, iv–d

(2) i–b, ii–a, iii–c, iv–d

(3) i–b, ii–a, iii–d, iv–c

(4) i–d, ii–c, iii–b, iv–a

89) In frog the ventricle opens into a sac like structure on the ventral side of heart is called :-

(1) Sinus venosus

(2) Renal portal vein

(3) Conus Arteriosus

(4) Dorsal Aorta

90) Read the following statements carefully and choose the correct option.
Statement-1 : In frog, a triangular structure called sinus venosus joins the right atrium.
Statement-2 : In frogs, the elimination of nitrogenous wastes is carried out by a well developed excretory system, which consist of a pair of kidneys, ureters, a cloaca and a urinary bladder.

(1) Both the statements (1) and (2) are correct.

(2) Statement (1) is correct but statement (2) is incorrect.

(3) Statement (2) is correct but statement (1) is incorrect.

(4) Both the statement (1) and (2) are incorrect.

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	1	3	3	4	4	2	3	3	3	4	4	2	2	4	3	2	2	2	4
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	1	3	1	4	2	3	1	2	3	4	1	2	3	1	2	1	1	3	3
Q.	41	42	43	44	45															
A.	4	4	1	3	1															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	2	2	3	4	2	3	4	4	4	1	3	3	2	4	3	3	2	3	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	2	1	1	1	1	3	3	2	1	1	3	3	3	2	2	2	2	4	1	1
Q.	86	87	88	89	90															
A.	3	1	2	2	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	3	1	4	3	3	2	2	4	1	3	4	3	1	2	3	2	2	3	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	1	2	4	2	3	3	4	3	2	2	1	3	3	3	3	4	1	2	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	2	2	3	1	2	2	3	2	4	1	4	2	2	3	2	3	3	4	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	4	3	2	3	2	3	2	4	2	4	3	4	3	3	1	3	2	1	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	1	4	2	2	4	2	3	3	3	1										

SOLUTIONS

PHYSICS

1) Let $F \propto P^x V^y T^z$
by substituting the following dimensions :
$[P] = [ML^{-1}T^{-2}][V] = [LT^{-1}]$, $[T] = [T]$
and comparing the dimension of both sides
$x = 1, y = 2, z = 2$, so $F = PV^2T^2$

2)

Given $F = \frac{\alpha}{\beta + \sqrt{d}}$, where, F = force, d = density, α = unknown, β = unknown, [α] = ?
This question is based on law of homogeneity.

Here, $[\beta] = [\sqrt{d}] = \left[M^{\frac{1}{2}}L^{-\frac{3}{2}}\right]$

$[F] = \frac{[\alpha]}{[\beta + \sqrt{d}]} \Rightarrow [\alpha] = [F]\,[\beta + \sqrt{d}]$

$= [M^1L^1T^{-2}]\left[M^{\frac{1}{2}}L^{-\frac{3}{2}}\right]$

$= \left[M^{\frac{3}{2}}L^{-\frac{1}{2}}T^{-2}\right]$

3) $V = L \times A = L \times \frac{\pi d^2}{4}$

Percentage error

$= \frac{\Delta V}{V} \times 100 = \left[\frac{\Delta L}{L} + \frac{2\Delta d}{d}\right] \times 100$

$= \left[\frac{0.1}{5.0} + \frac{2 \times 0.01}{2.00}\right] \times 100 = 3\%$

4)

1N = n_2 new system

1 kg m s^{-2} = n_2 $M_2L_2T_2^{-2}$

So $n_2 = 1\left[\frac{kg}{M_2}\right]\left[\frac{m}{L_2}\right]\left[\frac{s}{T_2}\right]^{-2}$

$= 1\left[\frac{kg}{10g}\right]\left[\frac{m}{10cm}\right]\left[\frac{s}{0.1s}\right]^{-2}$

$= 1\left[\frac{1000g}{10g}\right]\left[\frac{100cm}{10cm}\right]\left[\frac{1}{100}\right]$

= 10

5)

Density $\rho = \frac{m}{v} = \frac{m}{\pi r^2 \ell}$

So $\frac{\Delta\rho}{\rho} = \frac{\Delta m}{m} + 2\frac{\Delta r}{r} + \frac{\Delta \ell}{\ell}$

$$\rho\,\% = \left[\frac{0.003}{0.3} + 2 \times \frac{0.005}{0.5} + \frac{0.06}{6}\right] \times 100 = 4\%$$

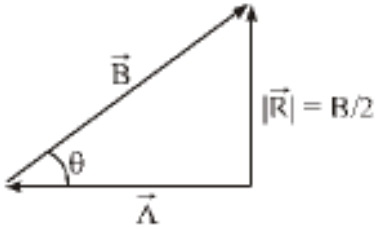
6)

$U = \frac{A\sqrt{x}}{x + B}$

$B = L \Rightarrow A = \frac{[ML^2T^{-2}]L}{\sqrt{L}}$

$A = ML^{5/2}T^{-2}$

$AB = ML^{7/2}T^{-2}$



7)

$\sin\theta = \frac{B/2}{B} = \frac{1}{2}$

$\theta = 30°$

Angle between $\vec{A}\&\vec{B}$ = = 180° – 30° = 150°

8)

Given:

$\vec{A} = 3\hat{i} + 4\hat{j},\ \vec{B} = 6\hat{i} + 8\hat{j}$

Magnitude:

$A = \sqrt{3^2 + 4^2} = 5$

$B = \sqrt{6^2 + 8^2} = 10$

Check each option

Option 1: $\vec{A} \times \vec{B} = 0$ $\vec{A} = 3\hat{i} + 4\hat{j}$ $\vec{B} = 6\hat{i} + 8\hat{j}$ $\vec{B} = 2\left(3\hat{i} + 4\hat{j}\right)$ $\vec{B} = 2\vec{A}$

$\theta = 0^\circ$

$\therefore \vec{A} \times \vec{B} = \vec{O}\ \left[\because \vec{A} \times \vec{B} = AB\,\mathrm{Sin}\,\theta\,\hat{n}\right]$

so the statement is true.

Option 2: $\frac{A}{B} = \frac{1}{2}$

$\frac{5}{10} = \frac{1}{2}$

True

Option3: $\vec{A} \cdot \vec{B} = 48$

$\vec{A} \cdot \vec{B} = 3 \cdot 6 + 4 \cdot 8 = 18 + 32 = 50$

Given value is 48, which is incorrect.

× False statement.
Option 4: A=5
Correct, we calculated A=5.
True

9)

∵ 10 (VSD) = 9 (MSD)
∴ 1VSD = $\frac{9}{10}$(MSD) = $\frac{9}{10}$× 0.1 cm = 0.09 cm
Least count L. C. = 1 MSD – 1 VSD
L. C. = 0.1 cm – 0.09 cm = 0.01 cm
Net Reading NR = MSR + VSR × LC – ZE
Net Reading NR = 1.3 cm + 2 × 0.01 cm – 0 cm
Net Reading NR = 1.3 cm + 0.02 cm – 0 cm = 1.32 cm

10) Pitch (p) = $\frac{1\,mm}{2}$ = 0.5 mm
Least count (LC) = $\frac{pitch}{No.of\,CSD}$
= $\frac{0.5\,mm}{100}$ = 0.005 mm
In fig (1)
There is a Negative zero error
ZE = – 4 LC = – 4 × 0.005 mm = –0.02 mm
In fig (2)
Main scale reading (MSR) = 4 pitch = 4 × 0.5 mm = 2 mm
Circular scale reading (CSR) = 55 LC = 55 × 0.005 mm = 0.275 mm
True Reading = 2 mm + 0.275 mm – (–0.02 mm) = 2.295 mm

11) Pitch = $\frac{1\ mm}{20}$ = 0.5 mm
L.C. = $\frac{Pitch}{N} = \frac{0.5}{50}$ = 0.01 mm
MSR = 3 mm
CSR = 35 × L.C. = 0.35 mm
Diameter = M.S.R + C.S.R – zero error
= 3 + 0.35 – (–0.03)
= 3.38 mm

12)

x/3	2x/3
20	60

$V_{avg} = \frac{x}{\frac{x}{60} + \frac{2x}{180}} = \frac{1}{\frac{1}{60} + \frac{1}{90}} = \frac{180}{3+2}$
$= \frac{180}{5} = 36$ Km/hr

13)

With constant positive acceleration, speed will increase when velocity is positive, speed will decrease, if velocity is negative.
Similarly, with constant negative acceleration, speed will increase, if velocity is negative and speed will decrease, if velocity is positive.
Hence, A → p, q, B → p, q, C → r, D → p, q.

14) For backward journey

$x = \frac{1}{2}(5)(1)^2 = \frac{5}{2}m$

So total distance $3x + x + x = 5x = \frac{25}{2}m$

15)

$u_{red} = 50 + 50 = 100$ km/h,
$a_{red} = 20 - 20 = 0$
$s_{red} = 100$

$\therefore t = \frac{100}{100} = 1$ hr.

$\therefore S_A = u_A t + \frac{1}{2}a_A t^2$

$= 50 \times 1 + \frac{1}{2} \times 20 \times (1)^2 = 60km$

16)

$u_{rel} = 50 - (-50) = 100$ m/s
$a_{rel} = g - g = 0 \quad \therefore v_{rel} = u_{rel} = 100$ m/s

17)

Compare with $y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$

$\tan\theta = 1 \Rightarrow \theta = 45°$

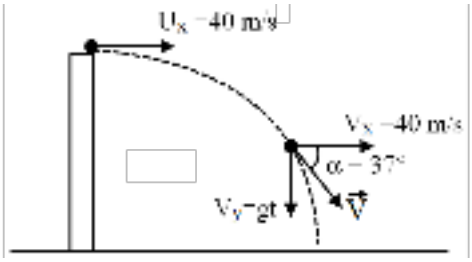
$\frac{g}{2u^2\cos^2\theta} = \frac{1}{80}$

$u\cos\theta = 20$ m/s

$u = 20\sqrt{2}$ m/s

$H_{max} = \frac{u^2\sin^2\theta}{2g} = \frac{800 \times \frac{1}{2}}{20} = 20m$

$R = \frac{u^2}{g} = \frac{800}{10} = 80$ m

18) 

$$\tan\alpha = \frac{V_y}{V_x}$$

$$\tan 37° = \frac{gt}{40}$$

$$\frac{3}{4} = \frac{10t}{40}$$

$$\boxed{t = 3\,sec}$$

19) $$\frac{dX}{\sqrt{X}} = \sqrt{\alpha}dt$$

$$\int_0^X X^{-1/2}dX = \sqrt{\alpha}\int_0^t dt$$

$$2\sqrt{X} = \sqrt{\alpha}t$$

$$X = \frac{\alpha t^2}{4}$$

20) $V_H = u\cos\theta = 6$

$V_v = \sqrt{v^2 - u^2\cos^2\theta} = 8$

$$t_1 = \frac{u\sin\theta - 8}{10}$$

$$t_2 = \frac{u\sin\theta + 8}{10}$$

$$t_2 - t_1 = \frac{8\times 2}{10} = 1.6s$$

21) Let velocity of particle at x = 3m is v m/s, then $\frac{v-10}{3} = 2 \Rightarrow v = 16$ m/s

The acceleration of particle at any instant is given by, $a = v\frac{dv}{dx}$

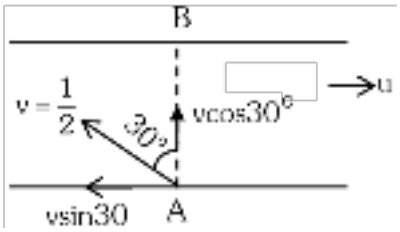
$\Rightarrow a = 16 \times 2 = 32\ m/s^2$

22)

$\tan\theta = \tan 45° + \tan 30°$

$$\tan\theta = 1 + \frac{1}{\sqrt{3}} = \left(\frac{\sqrt{3}+1}{\sqrt{3}}\right)$$

$$\theta = \tan^{-1}\left(\frac{\sqrt{3}+1}{\sqrt{3}}\right)$$

23) 

u = v sin30°

$u = \frac{1}{4}$

24)

$\because S = Ut + \frac{1}{2}at^2$

$\because -30 = 25t - \frac{1}{2} \times 10 \times t^2$

$\because t^2 - 5t - 6 = 0$

$\because t^2 - 6t + t - 6 = 0$

$\because (t - 6)(t + 1) = 0$

$\because t = 6$ sec. and $t = -1$ sec

So $t = 6$ sec

25) 

Time of flight = $\frac{2u\sin 30}{g} = \frac{2 \times 80 \times 1/2}{10} = 8$ sec

$\vec{V}_{av} = \frac{\vec{V}_{t=2} + \vec{V}_{t=6}}{2} = u\cos 30 = 80 \times \frac{\sqrt{3}}{2} = 40\sqrt{3}$ m/s

26)

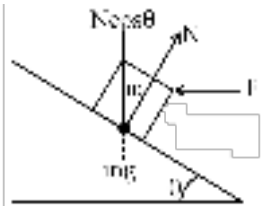
$T_{max} - mg = ma_{max}$

$600 - (40 \times 10) = 40 \times a_{max}$

$200 = 40a_{max}$

$a_{max} = \frac{200}{40} = 5 m/s^2$

27)



In vertical N cosθ = mg

$$N = \frac{mg}{\cos\theta}$$

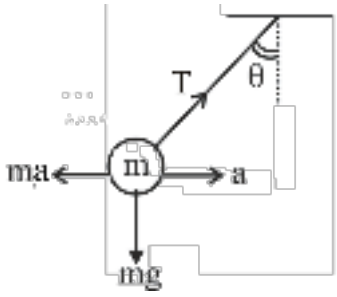
28)

$f_{max} = \mu_{mg} = 0.8 \times 4 \times 10 = 32N$

$t = 2$ sec $f = kt^2 = 2(2)^2 = 8N$

Applied force $f < f_{max}$ [block at rest]

force of friction will be 8 N



29)

T sin θ = ma ....(1)

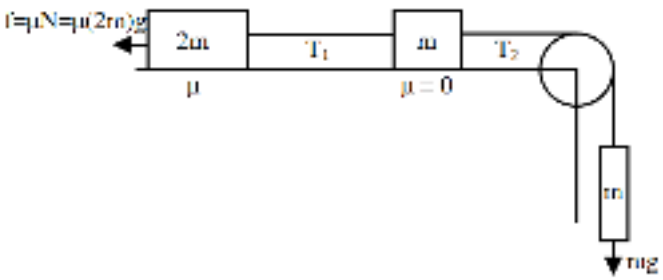
T cos θ = mg ....(2)

By equation (1) and (2)...

$$T = m\sqrt{a^2 + g^2}$$

$$T = 1\sqrt{\left(\frac{g}{\sqrt{3}}\right)^2 + g^2} \Rightarrow T = \sqrt{\frac{g^2}{3} + g^2}$$

$$T = \sqrt{\frac{4g^2}{3}} \Rightarrow T = \frac{2g}{\sqrt{3}}N$$



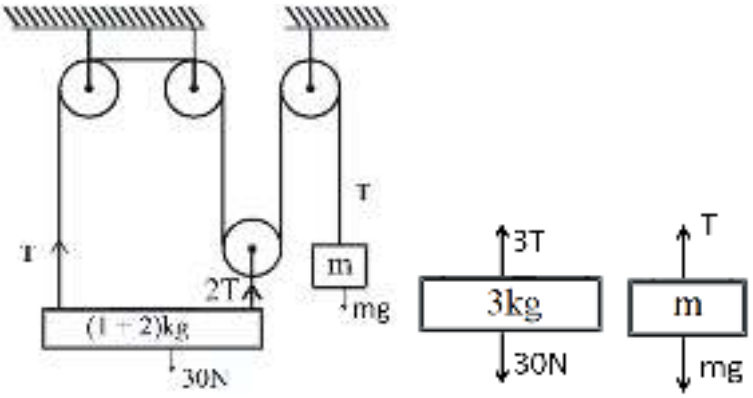
30)

$$a = \frac{F_{NET}\text{ along string}}{\text{total mass}}$$

$\because$ F_{NET} along string = 0

$\Rightarrow$ mg – μ(2m)g = 0

$\Rightarrow$ $\mu = \frac{1}{2}$



31)

3T = 30 .... (1)

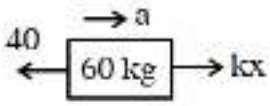
T = mg .... (2)

Put T in equation ------- (1)

3mg = 30

m = 1kg

32) $a = \frac{60-40}{60+40} = \frac{1}{5}\ m/s^2$



kx – 40 = 60 a

$100x = 40 + 60 \times \frac{1}{5} = 52$

x = 0.52 m

33) Net retarding force = Fcos 45° + μN

= Fcos 45° + μ(Fsin 45° + mg)

$= 20\sqrt{2} \times \frac{1}{\sqrt{2}} + 0.5\left(20\sqrt{2} \times \frac{1}{\sqrt{2}} + 5 \times 10\right)$

= 55 N

retardation = $\frac{F_{Net}}{m} = \frac{55}{5} = 11\ m/s^2$

velocity after 3 seconds.

V = u – at = 33 – 11 × 3 = 0

34)

F_{net} = mg sinθ – μmg cosθ

$\frac{F_{net}}{N} = \frac{1}{2} \Rightarrow \frac{mg\sin\theta - \mu mg\cos\theta}{mg\cos\theta} = \frac{1}{2}$

2 sinθ – 2 μcosθ = cosθ

2 sinθ – 2 × 0.5 cosθ = cosθ

⇒ sinθ = cosθ

θ = 45°

35)

$$a = \frac{nmv}{M} = \frac{10 \times 10 \times 10^{-3} \times 500}{2000}$$

$= 25 \times 10^{-3}$ m/s^2

$= 2.5$ cm/s^2

36)

(i) $a = \frac{2mg - mg}{3m} = \frac{g}{3}$

(ii) $a = \frac{2mg}{m} = 2g$

(iii) $a = \frac{2mg - mg}{m} = g$

(iv) $a = \frac{2mg}{3m} = \frac{2g}{3}$

37) $a = \left(\frac{m_2 - m_1}{m_1 + m_2}\right) g = 6 m/sec^2$ $a_{rel} = 12 m/s^2$

$S_{rel} = \frac{1}{2} a_{rel} t^2 \Rightarrow 6 = \frac{1}{2} \times 12 \times t^2 \Rightarrow t = 1$

38) $P = F \cdot v = m\left(v \cdot \frac{dv}{ds}\right) \cdot v$

$$\int_v^{2v} v^2 \cdot dv = \frac{P}{m} \int_0^s ds$$

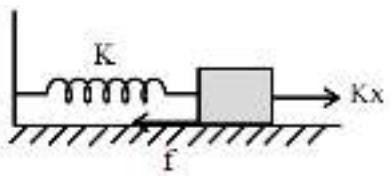
$$\left[\frac{v^3}{3}\right]_v^{2v} = \frac{PS}{m}$$

or $S = \frac{7mv^3}{3P}$

39) Work done = Gain in potential energy

Area under curve = mgh

$\Rightarrow \frac{1}{2} \times 11 \times 100 = 5 \times 10 \times h \Rightarrow h = 11m$

40) 

Block becomes to rest $F_{Net} = 0$

$f_{max} - Kx_{max} = 0$

$\mu mg = kx_{max}$

$X_{max} = \frac{\mu mg}{K}$

$$U_{max} = \frac{1}{2}Kx_{max}^2 = \frac{1}{2}K\frac{\mu^2 m^2 g^2}{K^2}$$

$$U_{max} = \frac{\mu^2 m^2 g^2}{2K}$$

41) At point 'C', the potential energy is minimum, hence it is a point of stable equilibrium.
Also, from E to F, the slope is negative i.e., $\frac{dU}{dr} < 0$
Hence, the force of interaction between the particles is repulsive between points E and F.

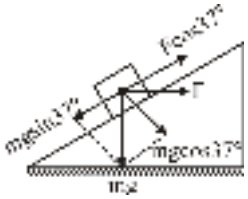
42) $P = \vec{F}.\vec{v}$ Tension is always perpendicular to velocity

43) $a \propto -x$
$a = -kx$
According to WET
$W = \Delta KE$
$Fx = \Delta KE$
$m(Kx)x = \Delta KE$
$\Rightarrow \Delta KE \propto x^2$
$\Rightarrow KE \propto V^2 \propto x^2$

44) Work done by the gravity (Wg) = mgh
$= 10^{-3} \times 10 \times 10^3 = 10$ J
By work–energy theorem
$= Wg + Wres = \Delta KE$
$10 + W_{res} = \frac{1}{2} \times 10^{-3} \times (50)^2$
$W_{res} = -8.75$ J

45)

As velocity is constant
$F\cos 37° = mg\sin 37°$
So WD by F
$W = (F \cos 37°)d$



$= (mg \sin 37°)d$
$= 50 \times \frac{3}{5} \times 2$
$= 60$ J

CHEMISTRY

46) $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

11 g

$n_{CO_2} = \frac{11}{44} = 0.25$

So $n_{CH_4} = 0.25$

47) (A) $n_{CH_4} = \frac{16}{16} = 1$

Moles of H atoms = $1 \times 4 = 4$

(B) $n_{H_2} = \frac{1}{2}$

$V_{H_2(g)}$ at STP $= \frac{1}{2} \times 22.7 = 11.35$ L

(C) Mass of $N_2 = 1 \times 28 = 28$ g

(D) Mass of $SO_2 = 0.5 \times 64 = 32$ g

48) $3BaCl_2(aq.) + 2Na_3PO_4(aq.) \rightarrow Ba_3(PO_4)_2 + 6NaCl$

moles 5 2

L.R. is Na_3PO_4

$\frac{n_{Na_3PO_4}}{2} - \frac{n_{Ba_3(PO_4)_2}}{1}$

$n_{Ba_3(PO_4)_2} = 1$

49) No. of Fe atoms = $2.8 \times \frac{0.34}{100} \times \frac{6 \times 10^{23}}{56}$

$= 1.02 \times 10^{16}$

50)

NCERT, Pg. # 12

51) $N_2 + 3H_2 \rightarrow 2NH_3$

(moles $\times 10^3$) $\frac{2.8}{2.8}$ $\frac{2}{2}$

$= 0.1$ 1

L.R. is N_2

$n_{NH_3} = 0.2 \times 10^3$

$W_{NH_3} = 0.2 \times 10^3 \times 17$

= 3.4 kg

52) (a) $n_{O_3} = \frac{16}{48}$

no. of atoms $= \frac{1}{3} \times 3 \times N_A$

(b) $n_{CO} = \frac{28}{28}$

no. of atoms = $1 \times 2 \times N_A$

(c) $n_{CO_2} = \frac{4.4}{44}$

no. of atoms = $0.1 \times 3 \times N_A$

(d) $n_{N_2O} = \frac{44}{44}$

no. of atoms = $1 \times 3 \times N_A$

53)

NCERT, Pg. # 197

54) $K'_C = \left(\frac{1}{K_C}\right)^2 = \left(\frac{1}{4.9 \times 10^{-2}}\right)^2$

$= \frac{10^4}{4.9 \times 4.9} = 416$

55)

NCERT, Pg. # 193

56)

$H_2(g)$	+	$I_2(g)$	⇌	2HI(g)
4.5		4.5		–
4.5 – 1.5		4.5 – 1.5		3

$K_C = \frac{(3)^2}{(3)(3)}$

57) $\Delta n_g = 1 - \left(\frac{3}{2}\right) = -\frac{1}{2}$

$\therefore K_P = K_C(RT)^{\Delta n_g}$

$\therefore K_P = K_C(RT)^{-\frac{1}{2}}$

$K_C = K_P(RT)^{\frac{1}{2}}$

$K_P = \frac{\alpha^2.P}{1-\alpha^2}$

$\alpha = \sqrt{\frac{K_P}{P}}$

58)

$PCl_5(g)$	⇌	$PCl_3(g)$	+	$Cl_2(g)$
1		–		–
1–α		α		α

$K_P = \left(\frac{\alpha}{1+\alpha} \cdot P\right)^2$

$K_P = \frac{\alpha^2 \cdot P}{1}$

$$\alpha = \sqrt{\frac{K_P}{P}}$$

59)

	A	+	B	⇌	C	+	D
	1		1		1		1
at eq$^{n.}$	1–x		1–x		1+x		1+x

$$K_C = \frac{(1+x)^2}{(1-x)^2}$$

$$9 = \frac{(1+x)^2}{(1-x)^2}$$

$$3 = \frac{1+x}{1-x}$$

3 – 3x = 1 + x

2 = 4x

$x = \frac{1}{2} = 1.5$ M

[D] = 1.5 M

60) OLD NCERT-XI, Pg. # 219

61) $pH = \frac{1}{2}(pK_w + pK_a + \log C)$

$pH = \frac{1}{2}\left(14 + 5 + \log 10^{-2}\right)$

= 8.5

62) OLD NCERT-XI, Pg. # 222

63) $\Delta H = \Delta U + \Delta n_g RT$

64) $\Delta_r H = [\Delta_f H(CO_2) + 4\Delta_f H(HCl)] - [\Delta_f H(CCl_4) + 2\Delta_f H(H_2O)]$

65)

NCERT-XI, Pg. # 162

66) $\Delta S = \frac{\Delta H}{T_{B.P.}}$

$75 = \frac{30 \times 1000}{T}$

T = 400 K

67) $Fe_2O_3(s) \rightarrow 2Fe(s) + \frac{3}{2}O_2(g);\ \Delta H = +822 kJ/mol$

$3C(s) + \frac{3}{2}O_2(g) \rightarrow 3CO(g);\ \Delta H = -330 kJ/mol$

$+$

$Fe_2O_3(s) + 3C(s) \rightarrow 2Fe(s) + 3CO(g)$

$\Delta H = +822 - 330$

$= +492$ kJ

68) $W = -P_{ext.}\Delta V$

$= -1$ atm $(10 - 1)$L

$= -9$ atm L

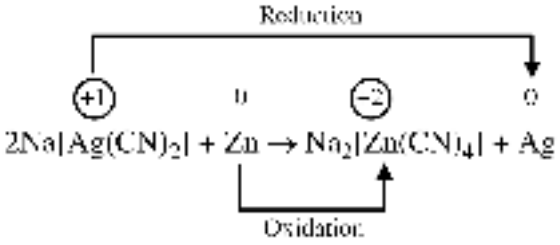
$= -900$ J

69) $Zr_3(PO_4)_4(s) \rightleftharpoons 3Zr^{+4} + 4PO_4^{3-}$

– 3s 4s

$K_{sp} = (3s)^3(4s)^4$

$= 6912s^7$

70) 

71) $HCl \rightarrow H^+ + Cl^-$

0.1

– 0.1 0.1

$HX \rightleftharpoons H^+ + X^-$

0.2 – –

0.2–x x x

$K_a = \frac{[H^+][X^-]}{[HX]}$

$10^{-5} = \frac{(0.1 + x)(x)}{(0.2 - x)}$

$x = 2 \times 10^{-5}$

$[X^-] = 2 \times 10^{-5}$ M

72) $C_xH_y + \left(x + \frac{y}{4}\right)O_2 \rightarrow xCO_2 + \frac{y}{2}H_2O(\ell)$

$V_{O_2} + V_{CO_2} = 340$

$500 - \left(80\left(x + \frac{y}{4}\right)\right) + 80x = 340$

$500 - 80x - 20y + 80x = 340$

$20y = 500 - 340$

$y = \frac{160}{20} = 8$

$V_{CO_2} = 80x$
240 = 80 x
x = 3

73)

NCERT, Pg. # 173

74) $NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$

at eq^m – $\frac{3}{2}$ $\frac{3}{2}$

$K_p = \left(\frac{3}{2}\right)\left(\frac{3}{2}\right) = \frac{9}{4} = 2.25$

75)

NCERT Pg. # 240

76)

NCERT, Pg. # 244

77) NCERT, Pg. # 247

78) $K_2Cr_2O_7 + Fe^{2+} \rightarrow Cr^{3+} + Fe^{3+}$
v.f. = 6 v.f. = 1
(0.2)(20)(6) = (M)(10)(1)
M = 2.4 M

79) Value of K_{sp} only depends upon temperature.

80) $[H^+] = \frac{4+6}{1000} = 10^{-2}$
pH = 2

81)

NCERT, Pg. # 213

82) $K_{sp} = 4s^3$ for AX_2
$4 \times 10^{-12} = 4s^3$
$s = 10^{-4}$
$K_{sp} = s^2$ for MX

$4 \times 10^{-12} = s^2$
$s = 2 \times 10^{-6}$ M
$$\frac{s(AX_2)}{s(MX)} = \frac{10^{-4}}{2 \times 10^{-6}} = 50$$

83)

NCERT, Pg. # 205

84)

$NH_4Cl + H_2O(\square) \rightleftharpoons NH_4OH + HCl$
$NH_4^+ + Cl^- + H_2O \rightleftharpoons NH_4OH + H^+ + Cl^-$
$NH_4^+ + H_2O(\ell) \rightleftharpoons NH_4OH + H^+$
$$K_h = \frac{K_w}{K_b} = \frac{10^{-14}}{2 \times 10^{-5}} = 0.5 \times 10^{-9}$$
$$= 5 \times 10^{-10}$$
$$pH = \frac{1}{2}(pK_w - pK_b - \log C)$$
$$= \frac{1}{2}\left(14 - 4.7 - \log 10^{-2}\right) = 5.65$$

85) $$K_{eq.} = \frac{K_f}{K_b} = \frac{10^3}{10^2} = 10$$
$\Delta_r G° = -2.303RT\log K_{eq.}$
$$\Delta_r G^o = -2.303 \times \frac{25}{3 \times 1000} \times 300 \log 10$$
= –5.75 kJ

86) $\Delta G = \Delta H - T\Delta S$
$$-50 = 51.4 - 300\frac{\Delta S}{1000}$$
$$-101.4 = \frac{-300}{1000}\Delta S$$
$$\Delta S = \frac{1014}{3}$$ J/mol-K
= 338 J/mol-K

87) For isothermal process of an ideal gas; $\Delta U = 0$

88)

NCERT, Pg. # 175

89)

NCERT, Pg. # 153

90) $CH_4(g) \rightarrow 4H(g) + C(g)$

$4BE(C - H) = x$

$BE(C - H) = \frac{x}{4}$

$C_2H_6(g) \rightarrow 2C(g) + 6H(g)$

$BE(C - C) + 6BE(C - H) = y$

$BE(C - C) + 6\left(\frac{x}{4}\right) = y$

$$BE(C-C) = y - \frac{6x}{4}$$

$$= y - \frac{3x}{2}$$

BIOLOGY

91) NCERT-XI (old), Pg. # 4-5

92) NCERT-XI, Pg. # 14, 15

93)

NCERT-XI, Pg. # 11

94) NCERT-XI, Pg. # 90

95) NCERT (XI) Pg. # 13

96)

NCERT-XI, Pg. # 90, 91

97)

NCERT-XI, Pg. # 13

98)

NCERT-XI, Pg. # 14

99) NCERT-XI, Pg # 12, 13

100)

NCERT-XI, Pg. # 90, 91, 13

101)

NCERT-XI, Pg. # 14, 15, 16

102)

NCERT-XI, Pg. # 14

103) NCERT Page nos : 14 to 16

104)

NCERT-XI, Pg. # 32

105)

NCERT-XI, Pg. # 28

106)

NCERT-XI, Pg. # 6

107)

NCERT-XI, Pg. # 17

108) NCERT-XI, Pg. # 17

109)

NCERT-XI, Pg. # 18

110)

NCERT-XI, Pg. # 17, 18

111)

NCERT-XI, Pg. # 20

112)

NCERT-XI, Pg. # 20, 21

113)

NCERT-XI, Pg. # 26

114) NCERT-XI Pg.# 28

115) NCERT-XI, Pg. # 32

116)

The correct answer is **3. They all may be used as food or food supplement.**
Explanation: **Porphyra, Agaricus, Chlorella,** and **Spirulina** are all used as food or food supplements.
• **Porphyra** is a type of red algae used in certain cuisines.
• **Agaricus** is a genus of mushrooms, commonly known as edible mushrooms.
• **Chlorella(Non-motile green algae)** and **Spirulina(Blue green algae)** are consumed for their nutritional benefits, often as supplements.

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121) NCERT-XI, Pg. # 32, 33

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126) NCERT-XI Pg. # 30

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128) NCERT-XI Pg. # 32

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130) NCERT-XI, Pg.# 5

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137) NCERT (XI) Example

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145) NCERT-XII, Pg. # 46, Para 4.1.1

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162) NCERT-XI, Pg # 102, Para.-7.1.1

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171) NCERT-XI, Pg. # 83

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